

Archaeologist

14+3 · lineage · intellectual move · sequels

Tool-home teleop → Bridge/V2 → RH20T/RT-1 → DROID → OXE → Octo → OpenVLA DROP
→ π_0 /FAST/pi05_droid → BEHAVIOR'25-'26.

Spine: Same Franka · many scenes · protocol > vibes · Fig.10 before Fig.8 · 2-16-8 reactive · steal
protocol into BEHAVIOR

Intellectual move: flip OXE's many-embodiments axis → many authentic scenes on one research arm

Lineage — big picture

Where DROID sits in the chain

Tool-home teleop (DobbE, Visual Imitation Made Easy)

Bridge / BridgeV2 — WidowX scene scaling

RH20T / RoboSet / RT-1 — lab-scale teleop

DROID (2024) — same Franka, many buildings, protocolized diversity

OXE aggregation — DROID as a slice (~311 scenes pooled)

Octo (curated OXE diffusion GRP) / RT-X

OpenVLA: DROID @10% → DROP mid-train (action-token underfit)

π_0 9.1% · FAST ZS · π_{05_droid} → BEHAVIOR'25-'26

Before DROID — what existed

Scene count was the scarce axis

Bridge / BridgeV2

- WidowX low-cost teleop
- Scene scaling pioneer
- BridgeV2 $\approx$ 24 scenes
- Verb tail comparable

RH20T / RoboSet

- Lab-scale multi-task
- Controlled setups
- Limited scene entropy
- Strong for skill coverage

RT-1

- Large traj count
- $\sim$ 2 scenes (kitchen-ish)
- Language + RT backbone
- Embodiment-specific

Gap DROID fills

- Same public Franka stack
- Hundreds of real buildings
- Protocol anti-bias
- Calibration + 3 views

Intellectual move

Flip the diversity axis

OXE axis

Many embodiments / aggregated lab pools
Who the robot is

DROID axis

Many authentic scenes on one research-standard arm
Where it works — with calib + 3 views baked in

Why it mattered

Showed scene entropy under fixed embodiment is a first-class generalization lever — not just more hours on the easiest task, and not only multi-robot soup. Protocolized anti-bias collection made the claim falsifiable (Fig. 10).

DROID moment — 2024

76k / 350h / 564 / 52 · CC-BY · 18 replicas

564 / 52

scenes / buildings

86 / 50

verbs / collectors

13 / 18

labs / robots

15 Hz

Polymetis control

2-16-8

DP horizons

+22 / +17

ID / OOD co-train

Not a new foundation model — methodology + diversity protocol is the product; DP results are the stress test.

Into OXE — DROID as a slice

Municipal water gains a scene-diversity shard

OXE pools DROID among ~60 datasets / 22 embodiments / 1M+ traj.

Agg scene count ~311 — still below DROID's solo 564 authentic scenes. DROID becomes the open Franka in-the-wild prior inside the municipal mix.

Kept

Scene-diversity reputation · Franka eval substrate · calib/views

Diluted

Protocol knobs not always preserved in soup sampling

Risk

Treating OXE dump as substitute for DROID-style churn

Octo sequel

Curated OXE diffusion GRP

Octo trains a diffusion generalist on a curated OXE split (Language Table often dropped for repetition).

DROID co-train experiments in the paper compare against this style of OXE soup — not the raw 1.4M dump. Reviewers: say so.

Positive

Open diffusion GRP baseline · shared eval habits

Limit

Still not a substitute for matched-N scene ablations

Link

Paper's OXE arm $\approx$ Octo-split moral

OpenVLA — DROP mid-train

Action-token underfit · not “bad DROID”

DROID @ ~10% of mix → DROP mid-train when discrete AR action tokens underfit the shard.

Same paper lineage: Franka-DROID Wipe fine-tune @15 Hz still beats DP/Octo — data remains valuable under the right action representation.

+

Scene diversity confirmed as lever

—

Diverse $\neq$ easy under naïve bins / AR tokens

System

Monitor per-shard action loss · pre-define kill switch

Later

FAST / flow / π action reps revisit fit

π -series reunion

9.1% · FAST ZS · pi05_droid

π_0 open mix

Open 9.1% = OXE Magic Soup + Bridge + DROID — DROID returns as a named shard

π_0 -FAST

DROID ZS language generalist narrative on campus rooms — shared Franka substrate

$\pi_{0.5}$ / OpenPI

pi05_droid full FT on RLDS v1.0.1 $\approx$ 75k lang episodes — pin version

Moral

Later foundation models both depend on and sometimes drop DROID by capacity $\times$ action-rep

Toward BEHAVIOR'25-'26

Steal protocol into stage-rich household

BEHAVIOR needs policies that survive new houses / layouts / long horizons.

DROID's durable gift is not 2024 ResNet DP weights — it is scene-churn protocol + matched-N diversity ablation discipline.

Map

GUI → BDDL random · relocation ·
distractor jitter

Ablate

Fig. 10 discipline before “need more
hours”

Race

On $\pi_{0.5}$ / N1.7 — not paper DP

What later papers learned

+ / - / systems

Positive

- Scene diversity = first-class generalization lever
- Protocol transparency beats vibes
- Open Franka stack = shared eval substrate
- Calibration upgrades travel (App. G)

Negative

- Diverse $\neq$ easy under discrete AR tokens
- OpenVLA drop mid-train
- Naïve concat can hurt
- Pre-VLA heads age out fast

Systems

- FAST ZS campus rooms use Franka prior
- pi05_droid versioned FT path
- Mix monitoring + kill switches
- Action-rep $\times$ fit co-design

Archaeologist exam bite

Ancestor in the OXE→VLA→ π chain

“DROID is the in-the-wild Franka ancestor in the OXE→VLA→ π chain; remember both the co-train gains (+22/+17, Fig.10 diversity) and the drop/underfit sequel (OpenVLA).”

Lineage tattoo

BridgeV2 → DROID flip → OXE slice → Octo → OpenVLA DROP → π_0 9.1% / FAST / π_{05_droid} → BEHAVIOR

Numbers tattoo — pin these

Shared across roles · Nov 6 ready

76k

successful traj

350 h

teleop hours

564

scenes

52

buildings

86

task verbs

50

collectors

13 / 18

labs / robots

+22% ID

Fig. 8 lift

+17% OOD

Fig. 8 lift

7362

Fig. 10 matched N

2 / 16 / 8

obs/pred/act @15Hz

~40k ≠ 76k

co-train pool

Archaeologist — close

Takeaways + exam bite

1. DROID flips OXE's many-robots axis to many authentic scenes on one public Franka stack.
2. Lineage: Bridge → DROID → OXE → Octo → OpenVLA drop → π_0 /FAST/pi05_droid → BEHAVIOR.
3. Later papers learned: diversity is a lever; diverse $\neq$ easy under AR tokens; open Franka = ZS substrate.
4. Durable gift = protocol + ablation discipline — not 2024 ResNet DP weights.

EXAM BITE

In-the-wild Franka ancestor; cite co-train gains AND drop/underfit sequel.